Architecture Design Challenge

As part of getting to know you better, you will participate in a collaborative architecture design exercise. This is a chance for us to learn how you approach technical challenges, and for you to get a feel for the type of problems we solve at Nerdery.

You'll find a hypothetical problem set outlined below. Please review it, and come prepared to discuss your approach during your technical interview. You don't need to have a fully worked-out solution beforehand, but feel free to explore it as much as you'd like. During our discussion, we're interested in hearing about:

1. The high level approach to solving the problem - including application structure (e.g. routes, data flow and management, logical segmentation of the problem space)
2. Technology selection decisions (frameworks, libraries, etc)
3. Any given specific section of the scope in detail (e.g. Authorization flow, user management)

If you have any questions about the problem or the exercise, don't hesitate to reach out to your recruiter. We're happy to help!

Thank you for your interest in Nerdery!

OVERVIEW

The client is the state's department of natural resources. They are looking to crowdsource the surveying of local fauna through the development of a web application that allows their users to keep track of the wildlife that they see and provide information about what makes different animals similar or distinct to encourage education on the state's wildlife. They want to allow users to "collect" animals and highlight what makes animals similar or distinct by mapping animal properties (habitat, diet, means of travel, etc) across the set of animals that are supported, including letting users select their "team".

Consider the high level design of the application and be prepared to do a deep dive into any specific section.

Functional Requirements

USERS

- There are two user types that will be using the application initially:

    - Biologists, who are responsible for adding both new animal definitions and new distinctive animal properties to the application - these will typically be DNR employees or volunteers.
        - Biologist users will be able to access distinct sections of the application which allow animal definitions and animal attributes to be added, updated, and removed from the application.
    - State citizens, who will be adding entries of the animals they "collect" and who will be able to construct their "team" of animals.
    - Biologists can also collect their own animals and build a team if they so desire.
- Because this is a public application, there may be 10's of thousands of users in the system
- User administration within the application will come as a fast follow - it is safe to assume that user roles will be able to be managed outside of the application for MVP.
- Users need to be able to sign-up for an account and login to the applications
    - The DNR has existing technologies they use for managing user identities which can be extended to this application
    - User properties include:
        - First Name
        - Last Name
        - email Address (must be unique)

ANIMAL DEFINITIONS

- Animals contain several intrinsic properties
    - Common Name
    - Scientific Name
    - "Avatar" picture - an image of the "on file" photo of the animal
    - Collection "number" - a public ID which represents the animals index order in the animal definition collection.
    - Range - A list of counties within the state where the animal can commonly be found
    - Types - Up to two from the list of 15, examples include: Flying, Aquatic, Bug, Rural, Metro, etc
    - Stats
        - Stats include properties like "Strength", "Size", and "Speed"
        - Right now these are just used to differentiate animals, but they may be used to gamify the application in the future.
    - For MVP, there will be ~1000 animal definitions
- Animals can have special attributes associated to them
    - These are most often are shared between many animals, but are occasionally unique
    - Each attribute has intrinsic properties
        - Name
        - Description

        - Selectable Values - The list of values that might apply to the attribute for a given animal
    - Examples of potential attributes include things like habitat, protection status, and diet
    - In the future, animal attributes could be expanded to allow more specific types, or allow select able values to be mapped to special bonus "stats" (e.g. nocturnal gives a bonus to "Speed" at night)

ANIMAL COLLECTIONS

- Animal definitions are listed in the collections overview page
    - Animals in a user's collection are either "collected" or "uncollected"
    - Only the common name, collection number, range, and file photo are able to be viewed if the user has not "collected" the animal, with the rest of the information unlocked
    - The collection is ordered by the collection number
    - The collection can be searched to quickly find an animal
- Users can "collect" animals by recording an encounter that they had with them.
    - When recording an animal, they must:
        - Select the animal they saw
        - Provide the location they saw the animal
            - This can be done manually or via device location
        - Upload a picture of the animal for verification
            - Verification will not currently be done in the application - it is out of scope for this phase, but will come later
        - They may optionally
            - Give the animal a nickname
- Users can collect an animal multiple times
- Users can view the set of animals they have collected
    - These are the unique individuals that users have recorded.
    - The user's photo and nickname (if provided) are used instead of definition values
- Users can select up to 6 animals as their "team" to display on their home page of the application when logged in
    - In the future, users will be able to interact with each others teams, allowing a more gamified experience
    - In the future, teams will be able to be shared on social media and other platforms

Non-functional Requirements and Assumptions

- Assume the use of modern browsers - Chrome, Firefox, Safari, Edge.
- The application must be responsive, with designs optimizing for mobile and desktop screen sizes.
- Multiple languages must be supported to accommodate the state's varied population

- To comply with federal guidelines and the Americans with Disabilities Act (ADA), the application must be WCAG 2.1 AA compliant
- The DNR expects application will eventually grow to having 10s of thousands of users, with a concurrent user base of up to 2000 at max load
- Due to the nature of the use case, the DNR is interested in allowing users to "collect" animals when there is no internet connectivity